7.6.7 Show that $X = HZH$

* $H = \frac{\sqrt{2}}{2}\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$
* $\sigma_x = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$
* $\sigma_z = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}$

$$\frac{\sqrt{2}}{2}\left[\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}\begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}\right] = \frac{\sqrt{2}}{2}\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}$$

$$\frac{\sqrt{2}}{2}\left[\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}\frac{\sqrt{2}}{2}\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}\right] =$$

$$\frac{1}{2}\begin{vmatrix} 0 & 2 \\ 2 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

σ_x

7.3.4 Why does $|\langle e_1|\psi\rangle|^2 = \langle\psi|e_1\rangle\langle e_1|\psi\rangle$

$$\Rightarrow |\langle +|\psi\rangle|^2 = \langle +|\psi\rangle^*\langle +|\psi\rangle$$

$$\Rightarrow |\langle e_1|\psi\rangle|^2 = \langle e_1|\psi\rangle^*\langle e_1|\psi\rangle$$

$$\langle +|\psi\rangle^* = \langle\psi|+\rangle$$

$$\Rightarrow |\langle e_1|\psi\rangle|^2 = \langle\psi|e_1\rangle\langle e_1|\psi\rangle$$

7.6.3 Show by calculation that $XZ = -ZX$

$$\sigma_x = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \quad \sigma_z = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} = \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} \quad \underline{\underline{XZ}}$$

$$\Rightarrow \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} \quad \underline{\underline{ZX}}$$

7.6.4 → Show by calculation that $XY = -YX$

$ZY = -YZ$

* $\sigma_x = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$
* $\sigma_y = \begin{vmatrix} 0 & -i \\ i & 0 \end{vmatrix}$
* $\sigma_z = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}$

⇒ First $X \cdot Y$

$$\Rightarrow \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \begin{vmatrix} 0 & -i \\ i & 0 \end{vmatrix} = \begin{vmatrix} i & 0 \\ 0 & -i \end{vmatrix}$$

⇒ Now $Y \cdot X$

$$\Rightarrow \begin{vmatrix} 0 & -i \\ i & 0 \end{vmatrix} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = \begin{vmatrix} -i & 0 \\ 0 & i \end{vmatrix}$$

⇒ First $Z \cdot Y$

$$\Rightarrow \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \begin{vmatrix} 0 & -i \\ i & 0 \end{vmatrix} = \begin{vmatrix} 0 & -i \\ -i & 0 \end{vmatrix}$$

⇒ Now $Y \cdot Z$

$$\Rightarrow \begin{vmatrix} 0 & -i \\ i & 0 \end{vmatrix} \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} = \begin{vmatrix} 0 & i \\ i & 0 \end{vmatrix}$$

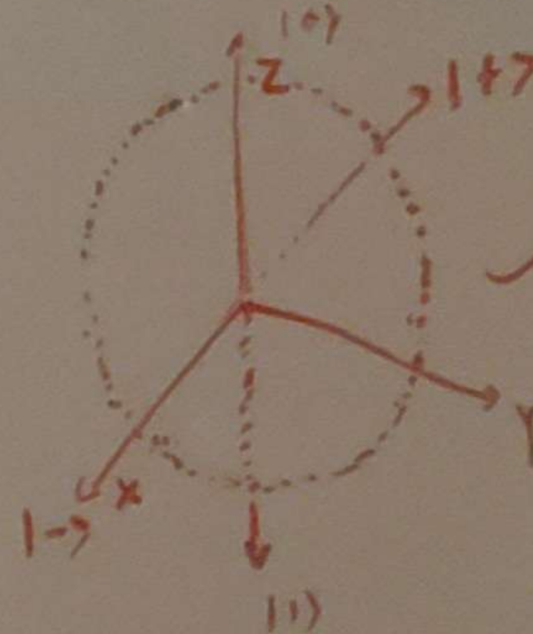
7.6.1 → What does X-Gate do to $|i\rangle$ and $|-i\rangle$?

* $\sigma_x = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$

** $\sigma_x |0\rangle = |1\rangle$

** $\sigma_x |1\rangle = |0\rangle$

*** Flips state from $|0\rangle$ to $|1\rangle$ or from $|1\rangle$ to $|0\rangle$



→ rotates by π around x-axis

$$\left\{ \begin{array}{l} X: +, - \\ Y: i, -i \\ Z: 0, 1 \end{array} \right\}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$$

↓

X-Gate flips $|i\rangle$ and $|-i\rangle$

↑

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

7.6.10 → 3x3 Matrix For S-Gate

* $S = R^{2}_{\frac{\pi}{2}}$, $\psi = \frac{\pi}{2}$

$$\Rightarrow \begin{vmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

7.6.12 3x3 Matrix For T Gate

$$T = R^{z}_{\frac{\pi}{4}}, \quad \varphi = \frac{\pi}{4}$$

$$\begin{vmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{vmatrix}$$